

# Resource Allocation in Multibeam LEO Satellite Systems Based on Beam Hopping and Frequency Reuse

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**Abstract**—The rapid expansion of low-Earth orbit (LEO) satellite networks exposes critical limitations in conventional resource allocation schemes, which are unable to simultaneously optimize spectral utilization and adapt to heterogeneous traffic patterns under extreme mobility, necessitating a joint beam-frequency dynamic coordination framework. To address the challenges of multibeam LEO systems, this article introduces a hybrid framework that synergizes adaptive beam activation patterns and spectrum reuse optimization, enhanced by a deep reinforcement learning (DRL)-driven coordination mechanism for resource allocation in time. By dynamically adjusting beam activation patterns and frequency allocation, the framework optimizes spatial-temporal resource utilization while mitigating co-channel interference. Angular-constrained multicriteria clustering achieves dynamic beam-user mapping with low-complexity adaptation for LEO mobility. The DRL-based component further coordinates multidimensional parameters, including transmit power and subchannel assignment, to balance throughput and latency under time-varying channel conditions. Extensive simulations validate the framework's capability to maintain high spectral efficiency and coverage performance across diverse scenarios, outperforming conventional static allocation methods. The results highlight its adaptability to dynamic traffic patterns and scalability for large-scale deployments, providing a robust foundation for next-generation LEO systems.

**Index Terms**—Adaptive dynamic multiple access, beam hopping, interference management, resource allocation, spectrum reuse.

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## I. INTRODUCTION

WITH the rapid development of modern communication technologies and the growing demand for global interconnectivity, low-Earth orbit (LEO) satellite communication systems have played an irreplaceable role in advancing the global informatization process [1], [2], [3]. Such systems can provide broad coverage enabled by multibeam architectures, flexible service areas through dynamic beam hopping, and strong adaptability via intelligent resource allocation strategies [4], [5]. However, the explosive growth in global Internet terminals has led to a shortage of wireless network spectrum resources [6]. In response to the escalating demand for mobile connectivity, the International Telecommunication Union's (ITU) latest report highlights the necessity of a tenfold enhancement in spectral efficiency to accommodate future wireless services, particularly in high-density user environments [7]. To meet this requirement, multibeam LEO satellite technology has emerged as a crucial solution for alleviating spectrum resource constraints [8], [9], [10]. The adoption of multibeam LEO satellite architectures is primarily driven by two major challenges in modern wireless networks: 1) the scarcity of spectrum resources, which cannot be resolved by simply increasing bandwidth to meet exponentially growing communication demands [11]; and 2) the highly uneven geographic distribution and traffic characteristics of user demand, which traditional static resource allocation models struggle to accommodate [12]. Therefore, multibeam LEO satellites, with their flexible coverage and resource scheduling capabilities, have become a key technology for optimizing resource utilization and enhancing communication efficiency [13], [14].

By dividing coverage areas into multiple smaller beam zones, each serving a specific geographic region, multibeam LEO satellites significantly enhance spectrum utilization efficiency and system capacity compared to traditional single-beam satellites [15]. Single-beam LEO satellites typically exhibit larger coverage footprints, which constrains their capacity to deliver high-quality services under limited spectrum resources. This architecture also lacks the flexibility to dynamically adapt to heterogeneous user demand patterns [16]. To overcome these limitations, beam hopping and frequency reuse strategy have emerged as key enablers for multibeam LEO constellations. By adopting frequency reuse techniques, multibeam satellites can significantly improve

spectrum efficiency [17]. Additionally, multibeam designs can provide higher bandwidth support for high-demand hot spot areas, better meeting the needs of modern communication services.

However, the combination of spatially heterogeneous user distribution and temporally fluctuating service requirements renders conventional multibeam configurations inadequate for maximizing spectral efficiency in dynamic LEO environments [18], [19]. Aiming at the static resource allocation limitations, beam hopping strategy has become a key solution in multibeam LEO systems [20], [21], [22]. Beam hopping dynamically adjusts beam coverage and service timing, effectively mitigating co-channel interference, improving system throughput, and adapting to the dynamic changes in user demand across different scenarios. For instance, in hot spot areas experiencing traffic surges, more beam resources can be temporarily concentrated to serve the region, while in low-demand areas, beam service frequency can be reduced, enabling flexible resource allocation. This on-demand approach not only optimizes resource efficiency but also alleviates imbalances in resource distribution across regions.

Frequency reuse is another core advantage of multibeam LEO satellite technology. By reusing the same frequency resources across different beams, the capacity of satellite communication systems can be significantly enhanced [23], [24]. However, frequency reuse also introduces interbeam interference, particularly at beam boundaries, which can adversely affect communication quality [25], [26], [27]. To address this issue, frequency reuse is cooptimized with beam hopping strategy, dynamically adjusting beam and power allocation strategies to reduce interference and optimize overall system performance [28], [29]. In the context of limited resources and diverse demands, the integration of frequency reuse and beam hopping, along with rational resource allocation, has become a critical research focus in satellite communication. Therefore, studying beam hopping coordination and frequency reuse optimization in multibeam LEO satellite systems is a critical pathway for advancing spectral efficiency in satellite communications. Which is also a fundamental requirement for achieving spatiotemporal resource utilization while mitigating interference in dynamic orbital environments.

#### A. Motivations and Contributions

Nowadays, there have been some researches on the dynamic resource allocation in multibeam LEO satellites, utilizing beam hopping and frequency reuse. Recent advances in multibeam LEO satellite resource management have established three principal research thrusts: 1) dynamic beam scheduling architectures [30], [31], [32]; 2) intelligent interference mitigation strategies [33], [34]; and 3) adaptive frequency reuse mechanisms [35], [36]. In particular, the work in [37] introduced a multiagent deep reinforcement learning (DRL) framework for joint beam pattern and bandwidth allocation, exploiting temporal, spatial, and spectral dimensions to optimize resource scheduling in static satellite environments. While this approach mitigates action space dimensionality

through distributed agents, it primarily targets geostationary (GEO) systems with static network topologies and uniform traffic patterns. Expanding on this work, Ran et al. [38] introduced a parameterized reinforcement learning framework for coordinated beam illumination and transmit power optimization in multibeam architectures. This multiagent approach mitigates dimensionality challenges arising from combinatorial action spaces while maintaining scalability in satellite–terrestrial networks. While this approach demonstrates improved throughput and energy efficiency in GEO scenarios with time-varying traffic, it inherits two critical limitations when applied to LEO constellations: First, the static orbital assumption neglects the continuous beam footprint shifts caused by LEO mobility, which require real-time angular adjustments to maintain coverage alignment. Second, the frequency reuse strategy allocates bandwidth without explicitly modeling interbeam interference escalation. Further extending multiagent DRL applications, the work in [39] employs digital twin-assisted frameworks to optimize beam hopping in multisatellite LEO systems, achieving improved load balancing through centralized coordination. However, this method assumes static interference environments and deterministic satellite trajectories, neglecting the dynamic interplay between rapid beam footprint shifts and escalating co-channel interference in dense frequency reuse scenarios.

In terms of interference issues, the work in [40] leverages spatial isolation patterns in non-GEO constellations to mitigate intersatellite interference while balancing traffic loads. Although this method demonstrates improved traffic satisfaction through static interference avoidance strategies, they inadequately address the dynamic interference coupling effects induced by LEO mobility and dense frequency reuse, where beam footprints continuously shift and edge sub-band collisions escalate. In response to challenges, such as dynamic spectrum resource variations and fluctuating terminal demands in multibeam systems, Li et al. [41] further introduced a joint beam hopping and adaptive dynamic multiple access design. The proposed solution combines beam hopping with cognitive spectrum sensing and nonorthogonal multiplexing, enabling dynamic adaptation to both terminal-level service requests and spectrum availability in integrated satellite–terrestrial networks. This approach demonstrates enhanced spectrum utilization efficiency while adapting to traffic demand fluctuations. Additionally, the work in [42] proposed a framework combining beam hopping with precoding techniques to meet the high demands of hot spot areas in GEO satellite systems. The study designed three beam illumination pattern optimization strategies—user demand fairness, neighboring beam penalty, and a low-complexity queue-based method—to tackle resource management and spectrum reuse challenges, significantly improving user demand satisfaction rates and system efficiency.

While prior studies have explored beam-time-frequency coordination strategies for multibeam LEO systems, critical limitations persist. While DRL and multiagent systems have enhanced spectral efficiency and throughput in resource scheduling, they remain challenged by three interconnected dynamics: 1) escalating interference due to mobility and

frequent handovers; 2) complex, time-varying network topologies with long transmission latencies; and 3) high-dimensional state spaces in rapidly changing communication environments. Furthermore, although existing studies partially address interference in frequency reuse architectures, further suppressing interference and ensuring Quality of Service (QoS) in high-density satellite deployments and multilayered interference scenarios remains a complex research problem. Consequently, it is necessary to explore more refined beam management techniques and more efficient interference suppression methods to enhance the adaptability and performance of multibeam LEO satellite systems across diverse environments, thereby unlocking the full potential of communication resources.

In this article, we aim to solve the following problems. First, develop adaptive beam hopping strategies that dynamically align with heterogeneous traffic demands and mitigate time-varying interference in high-mobility communication environments. Second, enhance spectral efficiency through frequency reuse optimization, reducing co-channel interference while maximizing spatial resource utilization. Third, optimize resource allocation via learning-driven algorithms to address complex network dynamics, including time-varying topologies and multidimensional resource constraints.

- 1) A novel dynamic frequency reuse and beam hopping joint optimization mechanism is proposed. In response to the satellite's high-speed mobility, an adaptive frequency reuse factor is designed for regions highly sensitive to interference and areas with high spectrum demand.
- 2) Meanwhile, based on user traffic requirements, a novel beam hopping pattern is generated to dynamically adjust the coverage area and service schedule of beams, ensuring adequate spectrum for hot spot regions.
- 3) A novel multidimensional resource allocation scheme based on DRL is introduced. A reward function that combines system throughput, interference levels, and user satisfaction is designed to ensure that user demands are met while minimizing interference and improving resource utilization.
- 4) The proposed framework is validated through extensive simulations, which evaluate its performance under different traffic conditions and interference scenarios. The results demonstrate that the joint optimization of dynamic frequency reuse and beam hopping significantly reduces interference, enhances spectrum efficiency, and improves overall system throughput.

## B. Organization

The remainder of this article is organized as follows. Section II constructs the system model for joint beam-frequency optimization, addressing the dynamic frequency reuse and beam hopping coordination in multibeam LEO satellite systems. Section III presents the DRL-based multidimensional resource allocation scheme, detailing the design of the reward function and its application for optimizing system performance. Section IV discusses the simulation setup

and results, demonstrating the effectiveness of the proposed framework in minimizing interference, improving resource utilization, and adapting to varying user demands and environmental conditions. Section V concludes by synthesizing the framework's contributions and delineating future extensions, including multilayered constellation coordination and innovative interference-aware beamforming techniques.

## II. SYSTEM MODEL AND PROBLEM FORMULATION

### A. System Model

Fig. 1 illustrates the system architecture, where phased array-enabled LEO satellites dynamically reconfigure multi beam footprints to serve nonuniformly distributed ground users under time-varying traffic and orbital kinematics. In this scenario, phased-array antennas dynamically reshape beam footprints (hexagons) to track mobile user clusters under orbital velocity  $\vec{v}_s$ . Each beam  $b_m$  is parameterized by its geographic center  $p_{b_m}$ , coverage radius  $r_b$ , and half-power beamwidth  $\Delta\theta_{-3dB}$ . The system consists of multiple LEO satellites each equipped with a phased array antenna capable of generating steerable Ka band beams. Beam footprints are dynamically adjusted, adapting to user mobility through real-time cluster tracking. The key components of the system are as follows: Let  $\mathcal{S} = \{s_1, s_2, \dots, s_N\}$  denote the set of satellites in the constellation, and let  $\mathcal{U} = \{u_1, u_2, \dots, u_N\}$  denote the set of ground users, distributed nonuniformly across a geographic region with time-varying traffic demands. Each satellite  $s$  can generate a set of beams  $\mathcal{B}_s = \{b_{s1}, b_{s2}, \dots, b_{sK}\}$ , where  $K$  denotes the maximum simultaneously activatable beams per satellite. The total number of beams in the constellation is given by  $\mathbb{B} = \sum_{s \in \mathcal{S}} K$ , depending on real-time traffic load per satellite. The geographical distribution of users is represented by their coordinates  $(x_n, y_n)$  for each user  $u_n$ .

The beams illuminate hexagonal cells on the ground, with coverage dynamically adjusted to align with user clusters and demand patterns, adapting to the rapid mobility inherent in LEO constellation. The system operates in the Ka-band with a total bandwidth  $W_{\text{tot}}$ , segmented into  $K$  orthogonal subchannels, each of bandwidth  $W_k = W_{\text{tot}}/K \forall k \in \{1, 2, \dots, K\}$ . The beam coverage problem aims to determine the beam positions  $p_{b_m} = (x_m, y_m)$ , beam radius  $r_b$  and half-power beamwidth  $\Delta\theta_{-3dB}$  for each beam  $b_m$ .

To optimize spectral efficiency and mitigate interference, a soft frequency reuse (SFR) scheme [43] is adapted for the LEO constellation. In our work, the bandwidth is divided into center sub-bands (allocated to beams serving high-demand central regions) and edge sub-bands (reused by adjacent beams with reduced transmit power to suppress co-channel interference). This allocation dynamically adapts to real-time traffic density and interference levels: center sub-bands maximize capacity in hot spot areas, while edge sub-bands prioritize interference suppression through spatial isolation and power shaping [44].

### B. Beam-Hopping Framework

A time-slotted beam activation framework is adopted, where the system timeline is divided into  $N$  fixed-duration slots, each



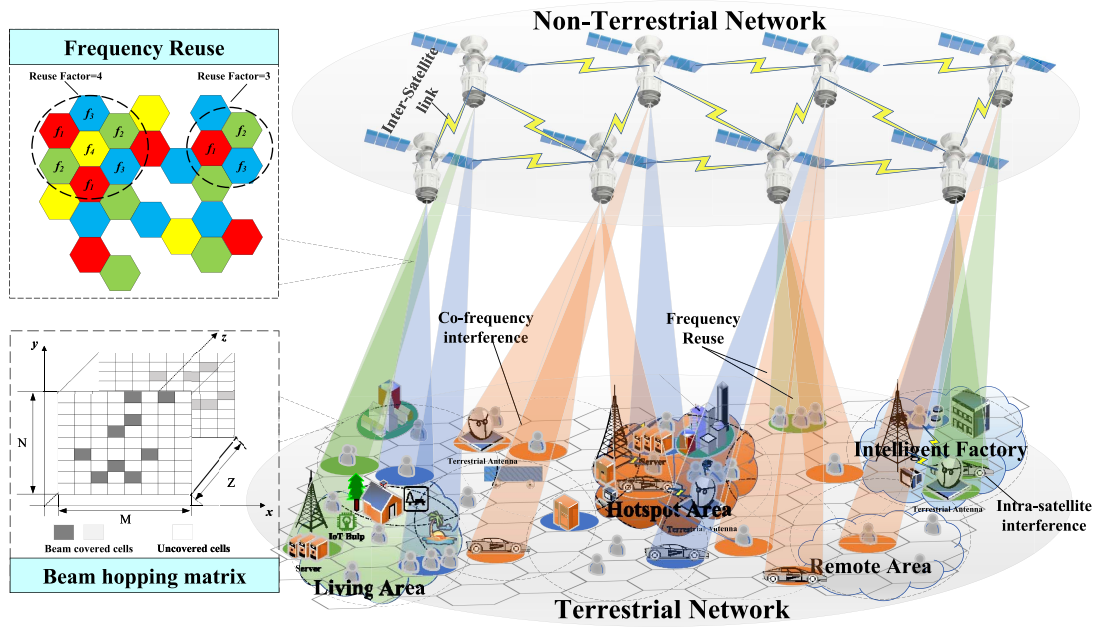


Fig. 1. Multibeam LEO satellite communication scenario.

spanning  $T_s$  milliseconds. Let  $\mathbf{A} = [a_{m,t}]_{B \times N}$  represent the beam activation matrix, which is given by

$$A = \begin{cases} 1, & \text{if beam } b_m \text{ is active in slot } t \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

The beam activation sequence dynamically adapts to the cumulative traffic demand  $D_b(t)$  within beam  $b$  at time  $t$ , which is expressed as

$$D_b(t) = \sum_{u=1}^{\mathcal{U}_b(t)} D_u(t) \quad (2)$$

where  $\mathcal{U}_b(t)$  is the set of users served by beam  $b$  at timeslot  $t$ .

The beam activation strategy employs a dynamic tessellation mechanism [45], where coverage cells are reconfigured based on real-time user density, reducing the required active beams. Specifically, users are clustered based on angular separation  $\beta_{u,v} \leq \Delta\theta_{-3dB}$ . This clustering is solved by a graph-based  $P$ -center optimization framework [46], which extends the classical  $P$ -center problem by incorporating angular constraints and dynamic user reassignment. Specifically, the number of active beams per slot is constrained by

$$\sum_{b \in \mathcal{B}_s} a_{b,t} \leq N_{b,\max} \quad \forall s \in \mathcal{S}, t \in \{1, 2, \dots, T\} \quad (3)$$

where  $N_{b,\max}$  denotes the interference-limited maximum number of concurrently activatable beams per satellite, determined by co-channel interference thresholds in frequency reuse scenarios.

Beam hopping operates by sequentially activating a set of directional beams to illuminate predefined geographical regions across  $N_T$  discrete time intervals, each spanning a fixed duration  $\Delta T$ . During these intervals, the system dynamically allocates transmit power and bandwidth resources throughout the entire simulation period  $T_{\text{sim}}$ , ensuring that resource

allocation adheres to a structured temporal framework. By partitioning the simulation timeline into fixed duration slots, the system maintains deterministic control over beam activation patterns, enabling precise coordination of power and bandwidth distribution across spatially heterogeneous user demands.

The beam hopping strategy is designed to dynamically allocate resources based on real-time user demand and interference conditions. First, the system collects real-time traffic demand  $D_b(t)$  and location information from users within the beam coverage area. User demand is aggregated into a pending traffic priority ( $\text{pending}_{c_j|t_i}$ ), weighted by historical unserved counts to balance fairness between high-demand and low-demand regions. Based on priority, the system activates the top  $N_{\text{beam}}$  high-demand beams.

### C. Channel Model and Gain Derivation

The high-orbital velocity of LEO satellites induces continuous spatial shifts in beam footprints, which necessitates real-time beam reconfiguration to sustain coverage alignment with ground users. The channel model explicitly integrates LEO-specific kinematics into the antenna gain formulation. To address these challenges, the antenna gain for user  $u_n$  under beam  $b_m$  at time  $t$  is modeled according to [47] as

$$G_{m,n}(t) = \eta \cdot \left( \frac{J_1(\alpha_{m,n}(t))}{2\alpha_{m,n}(t)} + 36 \cdot \frac{J_3(\alpha_{m,n}(t))}{\alpha_{m,n}^3(t)} \right)^2 \quad (4)$$

where  $\alpha_{m,n}(t) = (\theta_{m,n}(t)/\Delta\theta_{-3dB})$  is the normalized off-axis ratio, representing the user's angular offset relative to the beam's half-power beamwidth,  $\theta_{m,n}(t)$  denotes the angular deviation between beam  $b_m$  and user  $u_n$ ,  $\eta$  is the antenna aperture efficiency factor, and  $J_\omega(\cdot)$  is the  $\omega$ th order Bessel

function of the first kind ( $\omega=1,3$ ). The off-axis angle  $\theta_{m,n}(t)$  evolves with satellite motion

$$\theta_{m,n}(t) = \arccos\left(\frac{\vec{r}_{s,b}(t) \cdot \vec{r}_u}{|\vec{r}_{s,b}(t)| |\vec{r}_u|}\right) \quad (5)$$

where  $\vec{r}_{s,b}(t) \in \mathbb{R}^3$  defines the Earth-centered coordinates of satellite  $s$  emitting beam  $b$  at  $t$ , and  $\vec{r}_u$  is the user's position, both in Earth-centered coordinates. The time-varying distance  $d_{m,n}(t)$  is

$$d_{m,n}(t) = \sqrt{h_s^2 + 2h_s R_e + (R_e \sin \alpha_{m,n}(t))^2} - R_e \sin \alpha_{m,n}(t) \quad (6)$$

where  $h_s$  and  $R_e$ , respectively, denoting the satellite's orbital altitude (relative to Earth's surface) and Earth's mean equatorial radius. Leading to a free-space path loss [48]

$$L_{m,n}(t) = \left(\frac{\lambda}{4\pi d_{m,n}(t)}\right)^2 \quad (7)$$

where  $\lambda$  is the wavelength. Additional losses from atmospheric absorption and shadowing are modeled as a multiplicative factor  $\kappa(t)$ , yielding the total channel gain

$$h_{m,n}(t) = G_{m,n}(t) L_{m,n}(t) \kappa(t). \quad (8)$$

Co-channel interference occurs when adjacent beams share identical spectral resources. The signal-to-interference-plus-noise ratio (SINR) for user  $u_n$  associated with beam  $b_m$  is modeled as

$$\gamma_{m,n}(t) = \frac{P_m(t) h_{m,n}(t)}{\sum_{j \neq m} P_j(t) h_{j,n}(t) + N_0 W_k} \quad (9)$$

where  $P_m(t)$  is the transmit power of beam  $b_m$ ,  $h_{m,n}(t)$  is the antenna gain from beam  $b_m$  to user  $u_n$ ,  $L_{m,n}(t)$  is free-space path loss, which is calculated by (7).  $N_0$  is noise power spectral density, and  $\sum_{j \neq m} P_j(t) h_{j,n}(t)$  is aggregated interference arisen from other beams because of frequency reuse.

Power and bandwidth are allocated dynamically: Total power is distributed proportionally based on the demand of each beam, prioritizing high-traffic areas. As for beam allocation, an orthogonal edge-band scheme is used, dividing the frequency band into center bands (serving only center users) and edge sub-bands (reusable), reducing co-channel interference. To address the dual challenges of LEO orbital dynamics and spatially heterogeneous user density, a variable grid strategy is employed, dynamically generating beam coverage units using the graph-based  $P$ -center optimization framework to minimize the number of active beams. Beam centers are directed toward the geographic center of user clusters, with rapid adjustment via phased array antennas to ensure coverage efficiency. While hopping, adjacent beams are prohibited from using the same frequency band, and SFR is applied to divide edge and center bands, reducing intralayer interference.

#### D. Handover Integration

The transient nature of LEO satellite-ground links poses challenge in sustaining uninterrupted communication services. Owing to rapid orbital dynamics, the service continuity period

for a single satellite-user connection is typically less than 10 min [49]. To maintain persistent connectivity, terminals must seamlessly transition between successive satellite beams to minimize service degradation during link re-establishment. Existing solutions for transparent payload LEO satellites predominantly rely on schedule-based handover protocols [50], where centralized ground stations preallocate resources and coordinate transitions across thousands of terminals and satellites. While effective for static traffic patterns, these methods exhibit scalability bottlenecks in dense networks due to centralized decision latency and inflexible resource partitioning.

In contrast, our work focuses on regenerative payload architectures, which empower satellites with onboard processing capabilities to execute autonomous trigger-based handover. To formalize handover events, we introduce a binary handover matrix  $\mathbf{H} = [h_{o,u,t}]_{U \times T}$ , defined as

$$H = [h_{o,u,t}]_{U \times T} = \begin{cases} 1, & \text{if } \theta_{m,n}(t) > \theta_{b,\max} \\ 0, & \text{otherwise} \end{cases} \quad (10)$$

where  $\theta_{b,\max}$  is the maximum coverage angle of the satellite beam.

The handover process is intrinsically coupled with resource allocation through proactive subchannel reservation on target satellites and synchronized power tapering between serving and target beams. By integrating  $\mathbf{H}$  with the beam activation matrix  $\mathbf{A}$ , the system prioritizes resource allocation for transitioning users, leveraging real-time channel state information to minimize service disruption. This coordination ensures continuity during beam transitions while maintaining spectral efficiency across dynamically evolving network topologies.

#### E. Problem Formulation

We aim to maximize system utility while mitigating interference, handover overhead, and latency under the dynamic constraints of satellite mobility. The beam hopping schedule determines the spatial and temporal resource availability, serving as input to the resource allocation network, which in turn refines user-domain assignments and provides feedback to optimize future beam schedules. This interdependence enables a unified optimization of beam-domain and user-domain resources, addressing the LEO-specific challenges of frequent handovers and time-varying channel conditions.

The system utility is defined based on the achievable data rates of users, which are contingent on both beam activation and resource allocation decisions. For a user  $u$  assigned to subchannel  $k$  in beam  $b$  at time slot  $t$ , the instantaneous data rate is

$$r_{u,k,t} = W_k \log_2(1 + \gamma_{u,k,t}) \quad (11)$$

where  $W_k$  corresponds to the bandwidth allocated to subchannel  $k$ , and the SINR is defined in (9). The total data rate for user  $u$  over a beam hopping cycle of  $T$  slots is

$$R_u = \frac{1}{T} \sum_{t=1}^T \sum_{k=1}^K w_{u,k,t} r_{u,k,t} \quad (12)$$

where  $w_{u,k,t} \in \{0, 1\}$  indicates subchannel assignment to user  $u$ . To ensure proportional fairness across users, the system utility is

$$U = \sum_{u=1}^{\mathcal{U}} \log(R_u) \quad (13)$$

which balances throughput among users with varying channel conditions and demands.

However, in LEO systems, maximizing utility alone does not suffice due to mobility-induced challenges. Thus, the optimization objective incorporates penalties for interference, handovers, and latency, which is defined as

$$\max_{\mathbf{A}, \mathbf{W}, \mathbf{P}, \mathbf{H}} \left[ \sum_{u=1}^{\mathcal{U}} \log(R_u) - \lambda \sum_{t=1}^T \sum_{b=1}^{\mathcal{B}_t} \sum_{k=1}^K I_{b,k,t} - \mu \sum_{u=1}^{\mathcal{U}} \sum_{t=1}^T ho_{u,t} - \nu \sum_{u=1}^{\mathcal{U}} L_u \right] \quad (14)$$

where  $\mathbf{A}$  is the beam activation matrix,  $\mathbf{W} = [w_{u,k,t}]$  is the subchannel allocation matrix,  $\mathbf{P} = [P_{b,k,t}]$  is the power allocation matrix, and  $\mathbf{H} = [ho_{u,t}]$  is the handover decision matrix. The interference term  $I_{b,k,t}$ , which is from (10) representing co-channel interference from other active beams. The total handover cost is  $H_{\text{total}} = \sum_{u \in \mathcal{U}} \sum_{t \in T} ho_{u,t}$ , and latency  $L_u = Q_u/R_u$  is the ratio of queue length  $Q_u$  (accumulated from traffic demand  $D_u(t)$ ) to data rate, with weights  $\lambda$ ,  $\mu$ , and  $\nu$  balancing these terms.

The beam hopping component activates beams based on aggregated demand  $D_b(t)$ , which constrains resource allocation by defining the active beam set  $\mathcal{B}_t = \{b | a_{b,t} = 1\}$ , within which subchannels and power are assigned. The user domain resource allocation problem optimizes  $\mathbf{W}$ ,  $\mathbf{P}$ , and  $\mathbf{H}$  to meet user demands under these constraints, feeding performance metrics back to refine the beam schedule. This mechanism ensures that beam hopping outputs directly inform user domain resource allocation, while allocation outcomes shape subsequent beam decisions, achieving a holistic resource management strategy.

To address the high-dimensional state-action space inherent in joint optimization problem, a DRL framework leveraging an actor-critic architecture is designed. The agent iteratively refines its policy to maximize the long-term discounted reward formulated in (14), which integrates spectral efficiency, interference penalties, and QoS constraints unique to LEO dynamics.

The action space  $\text{act}_t$  encompasses beam activations  $\text{act}_t = [\text{act}_{1,t}, \dots, \text{act}_{\mathcal{B}_t,t}]$ , subchannel assignments  $W_t$ , power allocations  $P_t$ , and handover decisions  $ho_{u,t}$ . To manage dimensionality, the actor outputs: 1) a probability vector  $\pi_b$  for each beam's activation; 2) continuous power levels  $P_{b,k,t}$ ; and 3) binary handover decisions  $ho_{u,t}$ . Subchannel assignments within active beams are computed via a heuristic to reduce complexity.

The actor network processes  $\text{state}_t$  to produce action distributions: beam activations use a Bernoulli policy, power allocations use a Gaussian policy (mean  $\mu_p$ , variance  $\sigma_p^2$ ), and handovers use a logistic function. The critic network estimates

$V(\text{state}_t)$ , the expected reward from  $\text{state}_t$ , trained to minimize the temporal-difference error. The policy is updated using the advantage  $A(\text{state}_t, \text{act}_t) = R_t + \gamma V(\text{state}_{t+1}) - V(\text{state}_t)$  (with discount factor  $\gamma$ ), ensuring stable convergence via a proximal policy optimization (PPO) algorithm. Penalties in  $R_t$  discourage excessive handovers, while  $-\lambda I_t$  leverages SFR from the system model to optimize spectral efficiency.

Constraints ensure physical feasibility: beam activation is limited by

$$C1: \sum_{b=1}^{\mathcal{B}_t} a_{b,t} \leq N_{b,\max} \quad \forall s \in \mathcal{S}, t \in \{1, \dots, T\} \quad (15)$$

which ensures that the number of simultaneously active beams per satellite  $s$  does not exceed the interference-limited maximum number of concurrently activatable beams  $N_{b,\max}$ .

Subchannel allocation follows an orthogonal assignment principle to avoid intrabeam interference, which means a subchannel  $k$  in beam  $b$  can only be allocated to a user if the beam is active ( $a_{b,t} = 1$ ). It is constrained by

$$C2: \sum_{u=1}^{\mathcal{U}_b(t)} w_{u,k,t} \leq N_{b,\max} \cdot a_{b,t} \quad \forall b \in \mathcal{B}_t, k \in K \quad t \in \{1, \dots, T\}. \quad (16)$$

To mitigate co-channel interference in frequency reuse scenarios, the edge sub-bands are subject to spatial isolation constraints that prevent adjacent beams from simultaneously utilizing the same edge subchannel. Specifically, if beam  $b$  is allocated an edge subchannel  $k \in K_{\text{edge}}$ , its spatially adjacent beams  $b' \in \mathcal{N}_b$  should not activate the same subchannel in the same time slot  $t$ . This is formalized as

$$C3: w_{b,k,t} + \sum_{b'=1}^{\mathcal{N}_b} w_{b',k,t} \leq 1 \quad \forall b \in \mathcal{B}_t, k \in K_{\text{edge}} \quad t \in \{1, \dots, T\} \quad (17)$$

where  $\mathcal{N}_b$  denotes the set of beams adjacent to  $b$ . This constraint ensures that edge sub-bands are exclusively reused in nonadjacent beams, balancing spectral efficiency with interference suppression.

The total transmit power across all beams and subchannels must remain below the satellite's maximum power budget  $P_{\max}$ , preventing amplifier saturation and ensuring energy-efficient operation. It is constrained as

$$C4: \sum_{b=1}^{\mathcal{B}_t} \sum_{k=1}^K P_{b,k,t} \leq P_{\max}, \quad t \in \{1, \dots, T\}. \quad (18)$$

Each user  $u$  must achieve a minimum data rate  $R_{\min,u}$ , ensuring baseline service quality, which is given by

$$C5: R_u \geq R_{\min,u} \quad \forall u \in \mathcal{U} \quad (19)$$

which is derived from user-specific service level agreement requirements.

For latency-sensitive applications (e.g., real-time control, VoIP), the end-to-end delay  $L_u$  must not exceed a threshold

$L_{\max}$ , typically defined by application-layer protocols. It is given by

$$C6: L_u \leq L_{\max} \quad \forall u \in \mathcal{U}. \quad (20)$$

Regulatory compliance with the ITU radio regulations [51] is enforced through dynamic control of the equivalent power flux-density (EPFD), which quantifies the aggregate interference from LEO satellites to GEO networks. The EPFD impacting GEO station is quantified by

$$\text{EPFD}(t) = \frac{1}{4\pi} \sum_{i=1}^{N_{\text{LEO}}} \frac{P_i(t)G(\theta_i(t))G_{\text{GEO}}(\phi_i(t))}{d_i(t)^2} \quad (21)$$

where  $P_i(t)$  is the transmit power of the  $i$ th LEO satellite,  $G_i(\theta_i(t))$  is its antenna gain versus off-axis angle  $\theta_i$ ,  $G_{\text{GEO}}(\phi_i(t))$  is antenna gain of the GEO station versus off-axis angle  $\phi_i$ , and  $d_i(t)$  denotes the LEO-GEO Euclidean distance. The constraint is defined as

$$C7: \text{EPFD}_{\text{LEO} \rightarrow \text{GEO}}(t) \leq \text{epfd} \downarrow \quad (22)$$

where  $\text{EPFD}_{\max}$  is the maximum allowable EPFD for the relevant frequency band and service type. This ensures that the cumulative interference from LEO constellations remains within internationally mandated thresholds, preserving the operational integrity of GEO systems.

Constraints C1–C7 jointly govern the beam-frequency-power optimization space, ensuring adaptive tradeoffs among spectral efficiency, QoS heterogeneity, and ITU-compliant EPFD limits under LEO-specific mobility.

### III. JOINT BEAM HOPPING AND USER DOMAIN RESOURCE ALLOCATION FRAMEWORK

The interdependent optimization of beam hopping patterns and user-centric resource assignments in multibeam LEO constellations necessitates a kinematic-adaptive framework. This framework dynamically harmonizes time-varying traffic, interference constraints, and orbital dynamics through real-time frequency reuse adaptation and kinematic-driven beam adjustments. This section elaborates on a hierarchical optimization framework combining adaptive beam scheduling with DRL-based resource allocation. The framework addresses the challenges of high-dimensional state spaces, mixed discrete-continuous action spaces, and rapid channel variations inherent to LEO systems.

#### A. Traffic-Driven Beam Hopping With Adaptive Clustering

Beam activation is determined by real-time traffic demand and user distribution. Users are dynamically clustered into cells using a variable grid strategy: two users  $u_u, u_v$  are grouped if their angular separation  $\beta_{u,v} \leq \Delta\theta_{-3\text{dB}}$ . Given  $\mathcal{B}_t$  beams and  $\mathcal{U}_t$  users, a graph-based  $P$ -center optimization framework is introduced, which iteratively refines user groupings to minimize the maximum coverage radius across all beams while ensuring computational efficiency. This approach transforms the user grouping problem into a derivative of the  $P$ -center optimization framework, formulated as

$$\min_{\{\mathcal{P}_1, \dots, \mathcal{P}_B\}} \left\{ \max_{1 \leq b \leq B} \left\{ \min_{1 \leq u \leq U} d(\mathcal{P}_b, \mathcal{P}_u) \right\} \right\} \quad (23)$$

where  $\mathcal{P}_b$  denotes the geographical center of beam  $b$ , and  $d(\mathcal{P}_b, \mathcal{P}_u)$  represents the angular distance between user  $u$  and  $\mathcal{P}_b$ . The objective is to partition  $\mathcal{U}_t$  users into  $\mathcal{B}_t$  groups such that the largest coverage radius among all beams is minimized. The algorithm initiates by randomly selecting  $\mathcal{B}_t$  users as initial group centers. Each remaining user is assigned to the nearest group, forming an initial partition  $\mathbf{M} = \{\mathcal{M}_1, \dots, \mathcal{M}_B\}$ . For each group  $\mathcal{M}_b$ , the coverage radius  $r_b$  is calculated as the maximum distance from the group center  $\mathcal{P}_b$  to its constituent users

$$r_b = \max_{u \in \mathcal{M}_b} d(\mathcal{P}_b, \mathcal{P}_u). \quad (24)$$

The extremal group  $\mathcal{M}_{b_e}$ , characterized by the largest radius  $r_{b_e}$ , is identified as the primary target for refinement. Within  $\mathcal{M}_{b_e}$ , a subset  $I(\mathcal{M}_{b_e}) \subseteq \mathcal{M}_{b_e}$ , termed the binding subset, is identified such that the removal of any nonbinding user does not reduce  $r_{b_e}$ . For each user  $u \in I(\mathcal{M}_{b_e})$ , the algorithm evaluates whether reassigning  $u$  to a neighboring group  $\mathcal{M}_{b_r}$  would reduce the global maximum radius. Specifically, the user  $u$  is transferred from  $\mathcal{M}_{b_e}$  to  $\mathcal{M}_{b_r}$ , if the adjusted radii satisfy

$$\max (r_{b_e} - d(\mathcal{P}_{b_e}, u), r_{b_r} + d(\mathcal{P}_{b_r}, u)) < r_{b_e}. \quad (25)$$

The group centers  $\mathcal{P}_{b_e}$  and  $\mathcal{P}_{b_r}$  are then recalculated using the minimum circle coverage method to reflect the updated membership. This iterative reassignment process continues until no further reductions in  $r_{b_e}$  are achievable. The extremal group  $\mathcal{M}_{b_e}$  is subsequently removed from  $\mathbf{M}$ , reducing the problem to a  $\mathcal{B}_t - 1$  center optimization. The procedure repeats until all groups are processed, yielding a final partition where the maximum coverage radius is globally minimized.

The heuristic design achieves substantial complexity reduction through three optimized operations. User-cluster assignment costs  $\mathcal{O}(\mathcal{B}_t \mathcal{U})$  per iteration, as each user is mapped to the nearest center under angular constraints  $\beta_{u,v} \leq \Delta\theta_{-3\text{dB}}$ . Extremal cluster identification requires  $\mathcal{O}(\mathcal{B}_t)$  comparisons to locate  $r_{b_e}$ . Binding user reassignment dominates at  $\mathcal{O}(|I(\mathcal{M}_{b_e})| \cdot \mathcal{N}_b)$ , where  $|I(\mathcal{M}_{b_e})| \leq \mathcal{U}$  and  $\mathcal{N}_b$  denotes adjacent clusters. Crucially, hierarchical refinement converges in  $\mathcal{O}(\log \mathcal{U})$  iterations due to radius minimization in (25), yielding total complexity  $\mathcal{O}(\mathcal{B}_t \mathcal{U} \log \mathcal{U})$ . This is markedly lower than the  $\mathcal{O}(\mathcal{U}^3)$  complexity of exact  $P$ -center formulations [52], which relies on exhaustive pairwise distance computations and mixed-integer optimization. Consequently, the proposed heuristic achieves real-time feasibility for LEO constellations while preserving near-optimal clustering accuracy.

With user groups  $\mathbf{M}$  established, beam activation is determined by real-time traffic demand. The weighted pending traffic for cell  $c_j$  at timeslot  $t$  is determined by

$$\text{pending}_{c_j}(t) = \sum_{u=1}^{c_j} \left[ \left( D_u - \sum_{\tau=1}^{t-1} \text{served}_u(\tau) \right) \cdot \left( \tau + \frac{1 - \frac{1}{t-1} \sum_{\tau=1}^{t-1} \mathbb{I}_{\{\text{served}_u(\tau) > 0\}}}{1 + \frac{1}{t-1} \sum_{\tau=1}^{t-1} \mathbb{I}_{\{\text{served}_u(\tau) > 0\}}} \right) \right] \quad (26)$$



**Algorithm 1** Traffic-Driven Beam Hopping with Adaptive Clustering

**Input:** User set  $\mathcal{U}$ , satellite position  $\vec{r}_s(t)$ , beam count  $\mathcal{B}_s$ , and half-power beamwidth  $\Delta\theta_{-3dB}$

**Output:** Beam activation matrix  $\mathbf{A}$ , clustered groups  $\{\mathcal{M}_1, \dots, \mathcal{M}_B\}$

- 1: **Initialize:**
- 2: Randomly select  $B$  users as initial cluster centers:  $\{\mathcal{P}_1^{(0)}, \dots, \mathcal{P}_B^{(0)}\} \subset \mathcal{U}$
- 3: **Iterative Refinement:**
- 4: **while** not converged **do**
- 5:   Assign each user  $u$  to nearest cluster  $\mathcal{M}_b$  where  $\beta_{u, \mathcal{P}_b} \leq \Delta\theta_{-3dB}$
- 6:   **for** each cluster  $\mathcal{M}_b$  **do**
- 7:     Compute coverage radius:  $r_b = \max_{u \in \mathcal{M}_b} \beta_{u, \mathcal{P}_b}$  by Eq. (24)
- 8:   **end for**
- 9:   Identify extremal cluster:  $b_e = \arg \max r_b$
- 10:   Find binding subset  $I(\mathcal{M}_{b_e})$  where  $\forall u \in I(\mathcal{M}_{b_e}): r_{b_e} = \beta_{u, \mathcal{P}_{b_e}}$
- 11:   **for** each user  $u \in I(\mathcal{M}_{b_e})$  **do**
- 12:     **for** neighboring cluster  $\mathcal{M}_{b_r} \neq \mathcal{M}_{b_e}$  **do**
- 13:       **if** Eq. (25) is satisfied **then**
- 14:         Transfer  $u$  from  $\mathcal{M}_{b_e}$  to  $\mathcal{M}_{b_r}$
- 15:         Update  $\mathcal{P}_{b_e}$  and  $\mathcal{P}_{b_r}$  via minimum circle coverage
- 16:       **end if**
- 17:     **end for**
- 18:   **end for**
- 19: **end while**
- 20: **Beam Activation:**
- 21: **for** each timeslot  $t = 1$  to  $T$  **do**
- 22:   Calculate pending traffic  $\text{pending}_b(t)$  for each cluster via Eq. (26)
- 23:   Activate beams:  $a_{b,t} = \mathbb{I}[b \in \text{top}_{N_b}(\text{pending}_b(t))]$
- 24: **end for**

where  $\tau$  prioritizes underserved users. The top  $N_b$  cells with highest pending traffic are activated, and power is allocated proportionally

$$P_{b,t} = \frac{\text{pending}_{c_j}(t)}{\sum_{j=1}^{N_b} \text{pending}_{c_j}(t)} P_{\text{total}}. \quad (27)$$

This ensures high-demand regions receive adequate resources while maintaining proportional fairness across the coverage area.

### B. DRL-Based Multidimensional Resource Allocation

Within active beams, a PPO algorithm jointly optimizes subchannel assignment  $\mathbf{W} = [w_{u,k,t}]_{U \times K \times T}$  and power allocation  $\mathbf{P} = [P_{b,k,t}]_{B \times K \times T}$ . The state $_t$  includes: Real-time traffic demand  $D_b(t)$  from (2), channel gains  $h_{b,u}(t)$  from (8), queue lengths  $Q_u(t)$ , interference levels  $I_b(t)$  from (9), and handover indicators  $ho_u(t)$ . The action act $_t$  comprises: discrete subchannel assignments  $w_{b,k,t} \in \{0, 1\}$ , and continuous power adjustments  $\Delta P_{b,t} \in [-P_{\text{step}}, P_{\text{step}}]$  adhering to constraints.

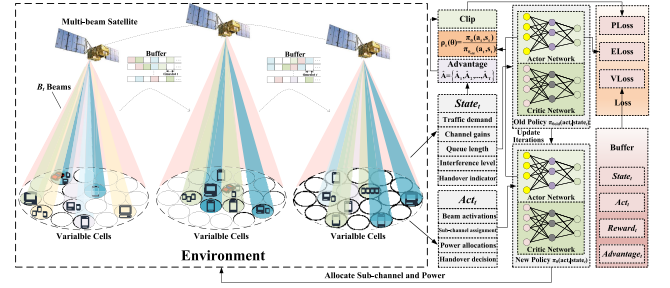


Fig. 2. Illustration of the multidimensional resource allocation.

Fig. 2 depicts the proposed multidimensional resource allocator, integrating beam hopping control, SFR-based spectrum partitioning, and DRL-driven parameters coordination.

The reward function is defined by (14), balancing throughput, fairness, and latency. The actor network employs a Gaussian policy for power allocation and a Bernoulli policy for subchannel assignment. The critic network estimates the state value  $V(\text{state}_t)$  using a temporal-difference objective

$$\mathcal{L}^{\text{critic}} = \mathbb{E}_t \left[ (r_t + \gamma V(\text{state}_{t+1}) - V(\text{state}_t))^2 \right] \quad (28)$$

where  $\gamma$  is the discount factor. The actor loss with clipped surrogate objective ensures policy stability

$$\mathcal{L}^{\text{actor}} = \mathbb{E}_t \left[ \min \left( \rho_t(\theta) \hat{A}_t, \text{clip}(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right] \quad (29)$$

where  $\rho_t(\theta) = \pi_\theta(\text{act}_t | \text{state}_t) / \pi_{\theta_{\text{old}}}(\text{act}_t | \text{state}_t)$  represents the policy probability ratio,  $\hat{A}_t$  denotes the advantage function, and  $\epsilon$  is set to 0.2.

The inherent dynamics of LEO constellations, marked by continuous orbital shifts and nonuniform user distribution, demand interference-aware beam management frameworks that simultaneously address co-channel interference mitigation and seamless handover operations. Central to this challenge is the interplay between beam hopping schedules, frequency reuse patterns, and predictive resource allocation, all of which must adapt in real-time to maintain service continuity and quality.

The online execution of Algorithm 2 per time slot comprises two components: 1) a forward pass through the actor-critic network and 2) resource-level bookkeeping. The actor and critic are two-layer perceptrons with  $\mathcal{I}$  trainable parameters, so the forward pass costs  $\mathcal{O}(\mathcal{I})$ , a constant once the model is deployed. Subsequent sampling of subchannel assignments and power updates touches at most  $\mathcal{O}(\mathcal{B}_t)$  active beams and  $K$  subchannels, incurring  $\mathcal{O}(\mathcal{B}_t K)$  time, while SINR computation and reward evaluation over the scheduled users add  $\mathcal{O}(UK)$ . Hence, the worst-case online complexity is  $\mathcal{O}(\mathcal{I} + \mathcal{B}_t K + UK)$ . During training, the PPO updating executes one backward pass for each batch of  $\mathcal{S}$  state-action pairs, costing  $\mathcal{O}(S\mathcal{I})$ . Combined with per-sample environment interactions, one epoch requires  $\mathcal{O}(S\mathcal{I} + \mathcal{S}(\mathcal{B}_t K + UK))$ . Both the online and training costs scale linearly with the cardinalities of beams, subchannels, and users, which confirms the practicality of the proposed multidimensional allocator for real-time LEO deployment.



**Algorithm 2** DRL-Based Multidimensional Resource Allocation With SFR

**Input:** Active beam set  $\mathcal{B}_t$ , channel gains  $h_{b,u}(t)$ , interference levels  $I_b(t)$ , queue states  $Q_u(t)$ , total bandwidth  $W_{tot}$ , and edge user density  $\alpha_b^{\text{edge}}(t)$

**Output:** Sub-channel assignment  $W_t$ , power allocation  $\mathbf{P}_t$ , SFR pattern  $\mathcal{F}_t$

- 1: Initialize agent with:
- 2: Actor network  $\pi_\theta: \mathcal{S} \rightarrow \mathcal{A}$
- 3: Critic network  $V_\phi: \mathcal{S} \rightarrow \mathcal{R}$
- 4: **Training Loop:**
- 5: **for** episode = 1 **to**  $E_{\max}$  **do**
- 6: Observe  $state_t = \{D_u(t), h_{b,u}(t), Q_u(t), I_b(t), H_u(t)\}$
- 7: **SFR Configuration:**
- 8: **for** each active beam  $b \in \mathcal{B}_t$  **do**
- 9: Partition sub-channels:
- 10:  $K_b^{\text{edge}} = \lfloor \alpha_b^{\text{edge}}(t) \cdot K \rfloor$  from Eq. (30)
- 11:  $K_b^{\text{center}} = K - K_b^{\text{edge}}$
- 12: Assign edge sub-channels with spatial isolation constraint
- 13: **end for**
- 14: **Action Generation:**
- 15: **for** each active beam  $b \in \mathcal{B}_t$  **do**
- 16: Sample sub-channel assignment:
- 17:  $w_{b,k,t} \sim \text{Bernoulli}(\pi_\theta^w(s_t))$
- 18: Enforce C2 per beam
- 19: **Power Allocation:**
- 20: Sample power adjustment:
- 21:  $\Delta P_{b,k,t} \sim \mathcal{N}(\mu_\theta^p(s_t), (\sigma_\theta^p)^2(s_t))$
- 22: Apply SFR power shaping: by Eq. (31)
- 23: **if**  $k \in K_b^{\text{edge}}$  **then**
- 24:  $P_{b,k,t} = \text{clip}(\text{Eq.31}, 0, P_{\max}^{\text{edge}})$
- 25: **else**
- 26:  $P_{b,k,t} = \text{clip}(P_{b,k-1,t} + \Delta P_{b,k,t}, 0, P_{\max}^{\text{center}})$
- 27: **end if**
- 28: **end for**
- 29: Enforce global power constraint in C4 from Eq. (18)
- 30: **Performance Evaluation:**
- 31: Calculate rates  $r_{u,k,t}$  via Eq. (11)
- 32: Compute reward by Eq. (14)
- 33: **Network Update:**
- 34: Compute TD error  $\delta_t = r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t)$
- 35: Update critic via Eq. (28)
- 36: Update actor with Eq. (29)
- 37: **end for**

TABLE I  
SIMULATION PARAMETER CONFIGURATION

| Parameter (Unit)                             | Value                                               |
|----------------------------------------------|-----------------------------------------------------|
| Orbital altitude                             | 780 km                                              |
| Orbital inclination                          | 86.4°                                               |
| Satellites per plane                         | 11                                                  |
| Number of planes                             | 6                                                   |
| Antenna aperture efficiency                  | 0.7                                                 |
| Coverage area                                | [109°E, 28°N], 6° span                              |
| User density                                 | {200, 400, 600, 800, 1000} (users/km <sup>2</sup> ) |
| User distribution model                      | Poisson Point Distribution                          |
| Satellite max transmit power ( $P_{\max}$ )  | 29 dBW                                              |
| Per beam max transmit power ( $P_{b,\max}$ ) | 18.5 dBW                                            |
| Power allocation step ( $P_{\text{step}}$ )  | 7 dBW                                               |
| Minimum elevation angle                      | 20°                                                 |
| Time step                                    | 1 second                                            |
| Interference limit ( $epfd \downarrow$ )     | -161.25 dBW/(m <sup>2</sup> ·40 KHz) [53]           |
| Beam hopping period                          | 1280 second                                         |
| Path loss model                              | Free space loss                                     |
| Fast fading model                            | Complex Gaussian (Var=1)                            |
| Rain attenuation model                       | ITU-R P.341-7 [54]                                  |
| Cloud/fog attenuation model                  | ITU-R P.840-8 [55]                                  |
| DRL learning rate (actor)                    | 0.0003                                              |
| DRL learning rate (critic)                   | 0.001                                               |
| Interference penalty weight ( $\lambda$ )    | 0.6                                                 |
| Handover penalty weight ( $\mu$ )            | 0.4                                                 |
| Delay penalty weight ( $\nu$ )               | 0.1                                                 |

half-power beamwidth

$$\alpha_b^{\text{edge}}(t) = \frac{\sum_{u=1}^{\mathcal{U}_b^{\text{edge}}} D_u(t)}{\sum_{u=1}^{\mathcal{U}_b} D_u(t)} \quad (30)$$

where  $\mathcal{U}_b^{\text{edge}}$  denotes edge users within beam  $B$  at time  $t$ . Edge sub-bands are sequentially allocated to adjacent beams to minimize overlap, ensuring spatial isolation.

The effectiveness of SFR is further enhanced through power shaping, where transmit power for edge sub-bands is selectively attenuated based on interference measurements. For edge user  $u$  in beam  $b$ , the adjusted power  $P_{b,k,t}^{\text{edge}}$  to subchannel  $k \in K_b^{\text{edge}}$  is governed by

$$P_{b,k,t}^{\text{edge}} = P_{b,t} \cdot \left( 1 - \frac{I_{b,k,t}}{\sum_{j=1}^{\mathcal{N}_b} I_{j,k,t} + N_0 W_k} \right) \quad (31)$$

where  $I_{b,k,t}$  represents interference measured at beam  $b$  on subchannel  $k$ , derived from (9). This closed-loop power control ensures that edge transmissions prioritize interference-prone regions while maintaining sufficient signal quality, keeping a balance between capacity and interference suppression.

## IV. SIMULATION RESULTS

## A. Simulation Setup

The proposed framework is evaluated under a multibeam LEO satellite system with parameters listed in Table I. The simulations consider both uniform and nonuniform user distributions, dynamic beam hopping patterns, and interference-aware resource allocation. Key configurations include a 780 km orbit altitude, Ka-band operation, and a maximum of 12 beams per satellite. The DRL agent is trained over 300 epochs with a batch size of 300, and power constraints adhere to ITU regulations as in (22).

To address co-channel interference, particularly at beam boundaries where overlapping coverage exacerbates signal degradation, a dynamic SFR scheme is implemented. Building on the foundational SFR framework introduced in Section II-B, the total bandwidth  $W_{\text{tot}}$  is partitioned into center and edge sub-bands, but with adaptive ratios tailored to real-time user distribution. For beam  $b$ , the edge sub-band ratio  $\alpha_b^{\text{edge}}(t)$  is dynamically adjusted based on the spatial density of edge users, defined as those located beyond half the beam's

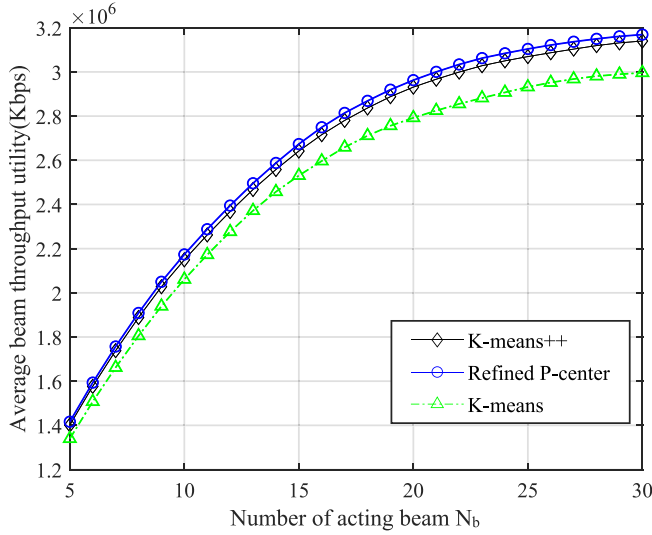


Fig. 3. Average beam throughput utility versus number of active beams.

### B. Beam Hopping Performance Analysis

The proposed refined *P*-center algorithm is compared with traditional *K*-means and *K*-means++ in terms of average beam throughput utility versus number of beams  $N_b$  in Fig. 3. As  $N_b$  increases from 5 to 30, throughput utility initially rises due to expanded resource allocation but saturates beyond  $N_b = 22$  (limited user demand and co-channel interference per (9)). The saturation beyond  $N_b = 22$  stems from two compounding effects: 1) diminishing marginal utility as beam granularity exceeds optimal spatial demand density, where additional beams fragment resources without new users; and 2) escalating co-channel interference from reduced spatial isolation between adjacent beams. At a fixed transmit power, spectral efficiency declines by 3.4% for highlighting the tradeoff between beam activation and interference. The refined *P*-center achieves 98.6%–98.9% of *K*-means++'s throughput while outperforming *K*-means, with faster convergence, aligning with LEO's real-time demands.

The simulation results in Fig. 4 demonstrate that the proposed clustering algorithm exhibits superior performance in uniform and nonuniform user distribution scenarios, achieving a 66.7%–76% and 70.2%–89.7% reduction, respectively, in required hopping cycles compared to conventional algorithms at 1000-user scale while maintaining over 92% final coverage efficiency, attributed to its synergistic optimization of spatial distribution and QoS demands through multidimensional feature clustering. In contrast, existing algorithms reveal significant performance divergence under clustered user patterns: the Max-Dur strategy suffers 58% coverage efficiency fluctuation in dense regions due to its over-reliance on temporal characteristics, while the Min-Dis approach, though initially efficient at low user densities, experiences 80% surge in hopping cycles during massive connectivity scenarios, highlighting the inherent limitations of single-dimensional optimization frameworks in addressing complex resource allocation challenges across heterogeneous satellite service environments.

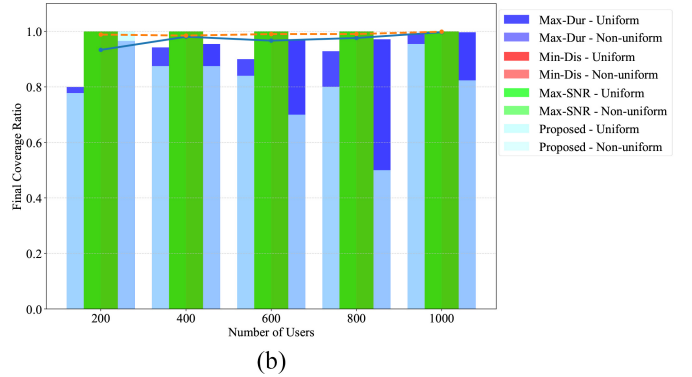
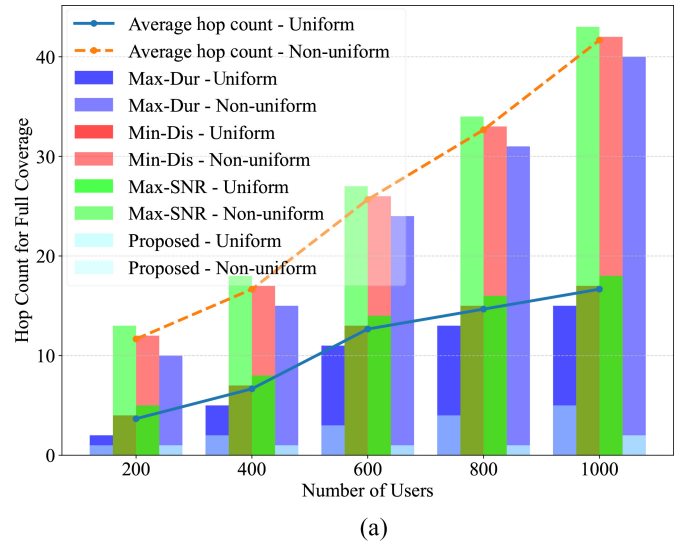


Fig. 4. Beam hopping performance in multibeam LEO systems. (a) Hopping count in different user density. (b) Coverage efficiency in different user density.

### C. Joint Resource Allocation Performance Analysis

As shown in Fig. 5, for uniform distribution, the reward curve exhibits a steady monotonic increase with minimal oscillations, achieving stable convergence ( $\Delta \text{reward} < 2.5\%$ ) beyond epoch 300, which indicates robust learning dynamics in homogeneous environments. Conversely, the nonuniform case reveals higher initial reward variance ( $\sigma^2 = 1.8$  versus 0.7 in uniform) during the exploration phase. The elevated initial reward variance in nonuniform scenarios directly reflects environmental complexity: clustered user distributions induce localized congestion, where the agent initially over-allocates beams to hot spot areas, starving peripheral regions and triggering handover penalties. This spatial imbalance amplifies exploration uncertainty until 150 epochs, when kinematic-aware beam hopping adapts coverage radii to demand heterogeneity. The stabilized reward plateau after 250 epochs in nonuniform distribution, suggests effective adaptation to spatially clustered user patterns through the integrated beam hopping mechanism. These differential convergence profiles quantitatively validate the algorithm's capacity to handle heterogeneous traffic loads while maintaining spectral efficiency in dynamic LEO environments.

As shown in Fig. 6, the SINR of the beam improves from -17.2 to -7.1 dB during training, driven by dynamic

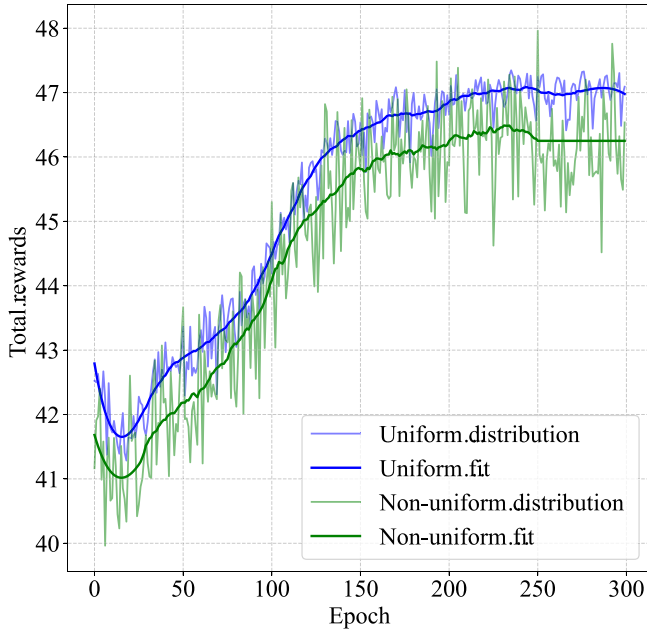


Fig. 5. Reward during DRL-based resource allocation.

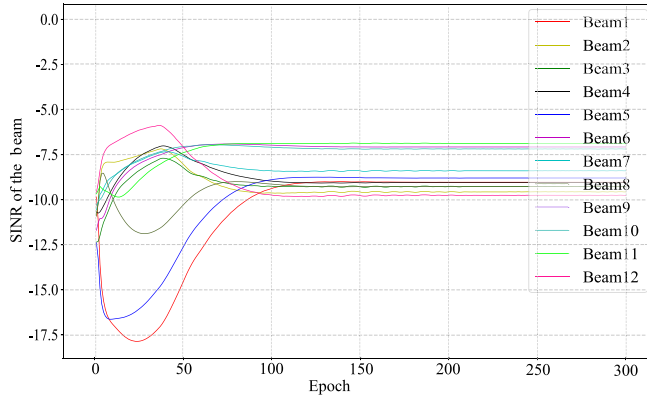


Fig. 6. SINR of per beam during the training.

power-gain coordination and interference-aware beam scheduling. The initial low-SINR stems from unoptimized power allocation in dense clusters, while the final value reflects effective co-channel interference mitigation via SFR. Notably, edge beams exhibit a 36.4% higher SINR improvement compared to central beams, attributed to prioritized power attenuation in overlapping regions. These results align with the joint optimization objective in (14), where penalty terms  $\lambda$  enforce interference suppression.

Beam-specific power [Fig. 7(a)] and gain (b) allocations exhibit heterogeneous patterns, reflecting spatial traffic variations. Beam 6 (hot spot region) sustains 15% higher average power than Beam 12 (low-demand area), while gain values range from 37 to 39 dBi, correlating with angular user density (5). The adaptive SFR mechanism ensures edge beams (e.g., Beam 5) operate at 8% lower power than central beams (e.g., Beam 6), reducing interference while maintaining coverage efficiency. This synergy between power and gain

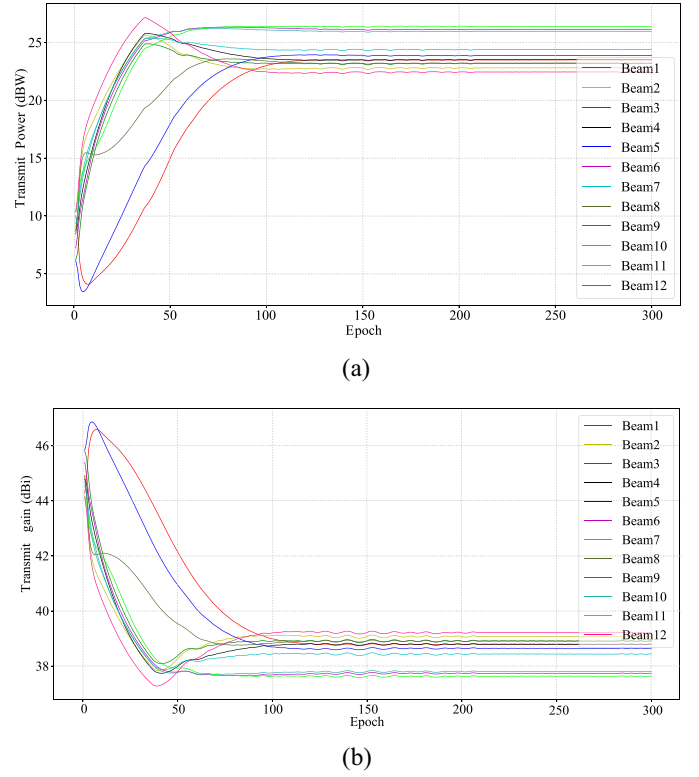


Fig. 7. Beam-specific power and gain allocation performance in multibeam LEO systems. (a) Transmit power during DRL-based resource allocation. (b) Transmit gain during DRL-based resource allocation.

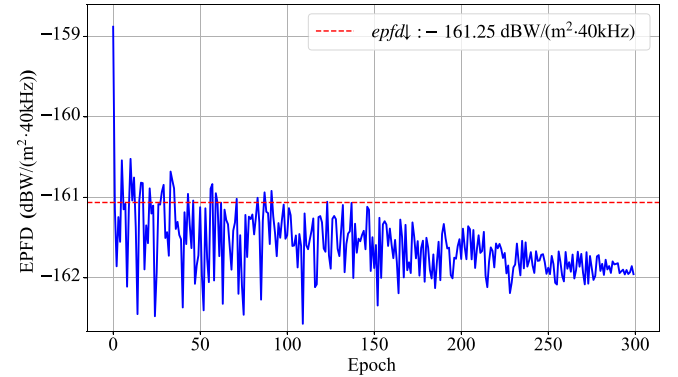


Fig. 8. Evolution of EPFD during optimization epochs.

optimization highlights the framework's ability to balance spectral efficiency and QoS requirements.

The variation of EPFD over training epochs in the multi-beam LEO satellite system under the proposed dynamic resource allocation framework is illustrated in Fig. 8. The EPFD limit  $epfd_{\downarrow}$  is set to  $-161.25$  dBW/(m<sup>2</sup>·40kHz), adhering to the ITU regulations to mitigate interference from the LEO satellite system to GEO satellite system [53]. Initial EPFD values at  $-158.8$  dBW/(m<sup>2</sup>·40kHz) converge toward the  $epfd_{\downarrow}$  after 300 training epochs, achieving a final mean EPFD of  $-161.5$  dBW/(m<sup>2</sup>·40kHz) with a safety margin of 0.9 dB. As shown in the simulation result, 95% of the operational intervals comply with the ITU mandate. This

behavior aligns with the theoretical guarantees of the DRL-driven optimization model, wherein reward shaping through SINR-weighted penalties and latency-aware resource partitioning ensures tradeoffs between throughput maximization and interference containment.

## V. CONCLUSION

This work presents a unified resource management framework for multibeam LEO systems, resolving the coupled challenges of dynamic interference suppression and spectral utilization through synergistic beam-frequency coordination. The proposed methodology bridges adaptive beam hopping, DRL-driven parameter optimization, and traffic-aware frequency reuse. By synergizing adaptive beam hopping with dynamic frequency reuse, the proposed methodology optimized spatial-temporal and spectrum resource allocation, enabling efficient coverage of heterogeneous traffic patterns. The incorporation of a DRL architecture further refined power and subchannel assignment coordination, balancing throughput demands with interference constraints across dynamically evolving network topologies.

The framework demonstrated robust performance in mitigating co-channel interference while maintaining high spectral utilization, even under dense beam activation scenarios. Its ability to prioritize high-demand regions through adaptive clustering and power shaping underscored its superiority over conventional static allocation strategies. The DRL-driven optimization ensured scalable adaptation to rapid channel variations and user mobility, critical for LEO systems operating in highly dynamic environments. Nevertheless, several inherent limitations merit further discussion. The proposed approach assumes quasi-stationary traffic and channel conditions within each scheduling interval, which is unable to fully capture the abrupt variations encountered in highly dynamic or adversarial environments; while the DRL-based algorithm achieves efficient resource coordination in simulations, its deployment in spaceborne platforms is constrained by onboard computational capabilities and inference latency, potentially affecting real-time responsiveness. Addressing these dimensions is imperative for enhancing the operational robustness and deployment readiness of the framework.

Future work will pursue three pivotal extensions: First, the proposed framework will be scaled to multilayered constellations with optical intersatellite links, which will require the development of novel cross-orbit interference coordination protocols to manage heterogeneous propagation delays and Doppler shifts. Second, quantum-resistant security mechanisms will be integrated into beam hopping control signals through the use of lattice-based cryptography, thereby addressing emerging threats to satellite command channels in contested environments. Third, a digital twin-enabled predictive resource allocation scheme will be developed, in which federated learning across ground stations anticipates traffic surges by correlating terrestrial Internet of Things data with satellite visibility windows. These directions will advance the framework toward ultrareliable, secure, and

latency-critical applications in 6G-integrated space-terrestrial infrastructures.

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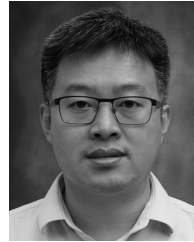
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